

WASTEWISE SCAN · REQUIREMENTS

WasteWise Scan requirements

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	WSC-REQ-001	Version	0.5
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wastewise-scan		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-02	Amish Chadha	Status from WSC-CAL-001 v0.3 for the constructable design (WSC-DDR-004); R6 not met; R12 reported against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); R4, R6, R11 and R12 restated per WSC-DDR-003; status from WSC-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 3 status from WSC-CAL-001; R12 redefined per WSC-DDR-001 D2; R13 points to the draft record in WSC-DDR-002
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

WasteWise Scan requirements

These are the requirements for the TRL 3 design. Status is from the calculation note WSC-CAL-001 v0.3 (tags in brackets), for the constructable design of WSC-DDR-004. R6 (mass) is not met on paper since the parts needed to build the scanner were added; R1, R7 and R13 are at risk, and R3 and R10 cannot be verified at TRL 3. R12 is reported against the USD 150 value-engineering target. Targets stay proposals for co-design with users. R12 was redefined by Amish's decision of 2026-09-25 (WSC-DDR-001, D2). At this revision R4, R6, R11 and R12 are restated to carry the decisions Amish accepted on 2026-09-25 (WSC-DDR-003): the backing step for clear items, the clip-on sun hood and the open-air calibration reading.

Table 1. Requirements.

ID	Requirement	Target	Status at TRL 3 (WSC-CAL-001)	Verification (TRL 3 or later)
R1	Identify common resins	PET, HDPE, PP, PS and PVC, plus "other" and "unknown"; 95 % or more correct on clean, dry, non-black items of 1 mm wall or more	At risk. The 1,650 nm band reaches only the short edge of the 1,650 to 1,750 nm region on standard InGaAs (centroid about 1,628 nm) [B2]; signal-to-noise ratio of 450 or more in shade [C4]	Band study on published reference spectra; later a labeled item set of 200 or more
R2	Fast result	1.5 s or less from button press to result	Met: about 0.33 s [E2]	Timing budget
R3	Fail safe on unreadable items	Black, dark, wet or very thin items read "unknown"; wrong resin shown for 2 % or fewer of all scans	Not verifiable at TRL 3. Black items fall well below the 0.08 reflectance threshold [C6]; the wrong-result rate needs item data	Confidence threshold study; later item test
R4	Work in sunlight	Same result indoors and in direct sun up to 100 klx when the shroud is pressed to the item; clear and translucent items are backed with the calibration cap, and the scanner refuses a result ("shade or back with cap") when its dark level shows light through an unbacked item (WSC-DDR-003 D8, D9)	Met by estimate: ambient error 0.13 % of signal on opaque items [D5] and 0.36 % on backed clear items [D6]; unbacked clear items in sun are refused [D7]	Stray light estimate; later indoor and sun test
R5	Last a shift	8 h or more at 4 scans per minute on one charge; USB-C charging	Met: about 22 h [F2]	Power budget
R6	One-handed and light	250 g or less including the sun hood; body no larger than 170 x 65 x 40 mm (6.7 x 2.6 x 1.6 in) plus shroud and clip-on sun hood	Not met: about 254 g, 274 g with cap, after the parts added for construction (was 209 g); 160 x 62 x 34 mm [A1, A3]. Options in WSC-DEC-001	Parametric model, then weighing
R7	Readable by anyone	Result as resin code, name, color and icon, legible at arm's length in direct sun; local language; no reading needed to use	At risk. With the clip-on sun hood (WSC-DDR-003 D10), about 5.2:1 where the hood shades the screen, fully shaded with the sun 50° or more off the screen normal on a side; about 1.6:1 with the sun near the normal or over the open side [I3, I4]	Contrast estimate; design review with users
R8	Local price grade	Shows a price grade from a table the user or cooperative edits on the device; works offline	Met by design	Design review
R9	Log scans	10,000 or more scans stored on the device with time, result, confidence and spectrum; export as CSV over USB; no cloud needed	Met: about 131,000 scans in an 8 MB partition [G1]	Storage calculation
R10	Rugged	Survives 1.2 m (4 ft) drop onto concrete; IP54 target; 0 to 45 °C	Not verifiable at TRL 3. Corner drop about 1,200 g; the cell strap holds	

ID	Requirement	Target	Status at TRL 3 (WSC-CAL-001)	Verification (TRL 3 or later)
			the cell against about 550 N with a margin of 1.3 [J1, J2, J4]; foam gasket between screwed shells	Design review, later drop and spray tests
R11	Stay calibrated	White reference check and an open-air window-crosstalk reading in 10 s or less using the calibration cap; warns when a check is overdue or drift exceeds 5 %	Met by design; 5 % drift after about 12.5 °C of temperature change [J3]. The crosstalk offset must stay within 0.5 % of itself between checks, which window dirt is likely to exceed [C7]	Design review
R12	Low cost	\$150 or less per unit is the volume target; the first prototype is accepted at about \$164 in parts (the \$163 accepted in WSC-DDR-001 D2 plus the \$1 sun hood of WSC-DDR-003 D10); no custom PCB for the first build	USD 167.50 [K2]: USD 17.50 over the USD 150 value-engineering target, USD 3.50 above the prototype figure accepted on 2026-09-25	Priced BOM
R13	Work with WasteWise-ml	Low-confidence scans hand off to WasteWise-ml; scan records use a shared, documented format	At risk. Hand-off designed; one shared record with a <code>pair_id</code> join decided on this side (WSC-DDR-002 v0.2, WSC-DDR-003 D12); not yet adopted by WasteWise-ml	Interface note with WasteWise-ml
R14	Eye-safe illumination	LED emission in the IEC 62471 exempt group; LEDs pulse only during a scan	Met by estimate: 3.9 W/m ² at 200 mm against 100 W/m ² ; retinal 3.8 % of the limit [H2, H3]	Emission estimate, then datasheet values

■ Assumptions

- "Clean and dry" means rinsed or wiped items without labels over the scan spot. Real waste is dirtier; the classifier must reject rather than guess.
- About 10 to 15 % of items in a mixed post-consumer stream are black or very dark (estimate; varies strongly by region and should be measured with users).
- The first prototype's accepted cost of about \$164 covers one scanner with its cap and sun hood; the \$150 volume target assumes quantity prices for LEDs and the photodiode (WSC-DDR-001, D2).
- Four scans per minute is a busy spot-check pace, not every item: most items are sorted by eye and the scanner settles doubtful ones.
- Resin prices vary by city and season; the device stores grades (for example A, B, C, reject) and a local price list, not global prices.